



MINING DEALER BEST PRACTICE

Setting Up the Metal Spray Robot Using a Cone Tool

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

It is necessary to keep distance and angle as accurate as possible during the spraying process to create a quality metal coat. Use of a robotic arm to control position and distance will provide the necessary accuracy.

To decrease the set-up time of the job, a cone tool is used to teach the dimensional specifications for the robot. Keep in mind that each type of arc spray wire needs different parameters as distance, voltage, etc. Use the information found in the Reuse and Salvage Guideline SEBF2120 “Thermal Spray Procedures for Engine Cylinder Block Top Surface” when spraying a cylinder block [SIS 2.0 \(cat.com\)](#).

2.0 Best Practice Description

The cone tool (Figure 1) is constructed of Nylamid (UHMW solid rod). This tool allows the teaching of programming points with greater accuracy, improving the quality of the coating.

It is necessary to install the cone into the arc shield of the arc spray gun, doing this keeps the center line (Figure 2) of the spray during the teaching of the points.

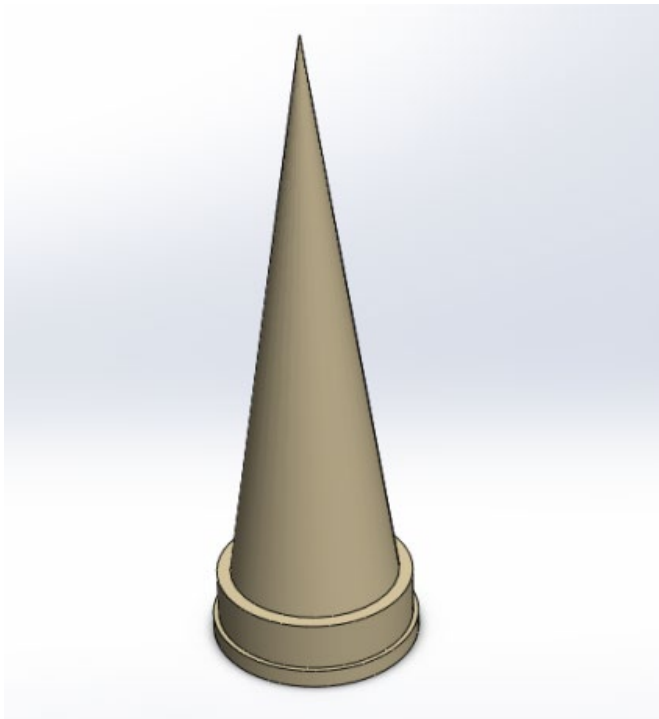


Figure 1

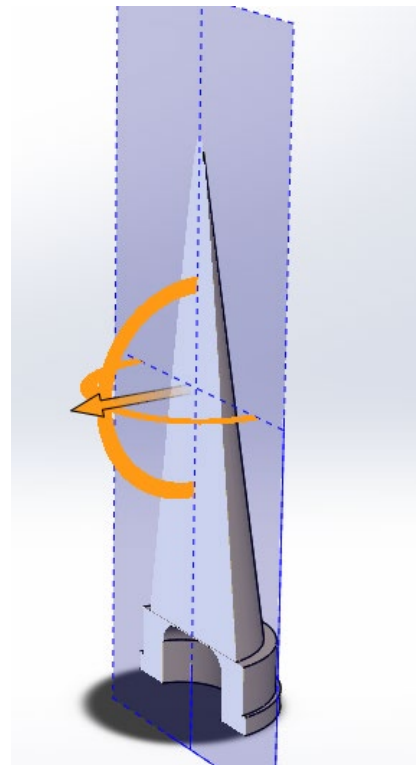


Figure 2

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3.0 Implementation Steps

3.1 Select the correct cone tool dimension, depending on the type of the wire being used in your application. The distance is found in the reuse and salvage guidelines. SEBF2120 “Thermal Spray Procedures for Engine Cylinder Block Top Surface”. For example, TAFA 75B or OERLIKON METCO 8400, a 5-inch-tall cone is utilized.

3.2 To assemble the cone tool in arc spray gun: Unscrew the arc shield from the arc spray gun (1), disassembly the air cap inserts from the arc shield (2), install the cone tool into the arc shield (3). Now screw the arc shield to the arc spray gun again.



Figure 2. Installing the cone tool into the arc spray gun

3.3 Teaching the robot using the cone tool: Once the cone tool is installed into the arc spray gun, the robot can be jogged to create the routine or path to metal spray the work piece.



Figure 3: Teaching the robot the spray path using a cone tool

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4.0 Benefits

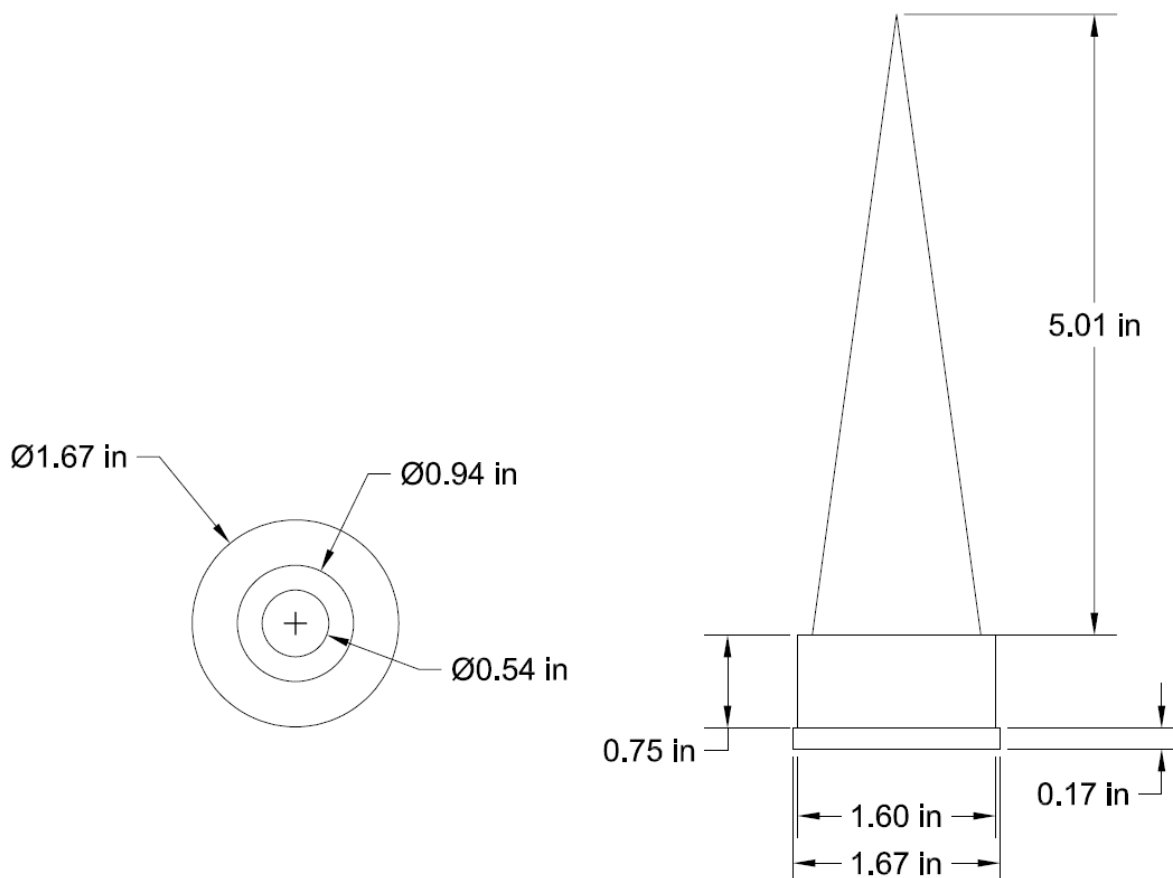
- Reduces the metal spray robot set-up time
- Allows salvaging the cylinder block liner bores of the C175 engines.
- Improves the quality of the coating

5.0 Resources Required

- Nylamid (UHMW solid rod) rod
- Horizontal lathe
- Metal spray equipment
- Industrial robot
- Manufacturing Price ~ 130USD

6.0 Supporting Attachments / References

- SEBF2120 “Thermal Spray Procedures for Engine Cylinder Block Top Surface”
[SIS 2.0 \(cat.com\)](#)
- Cone Tool Drawing - Units are Inches



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7.0 Related Best Practices

4.02-231002-2 Mounting Fixture for Metal Spray Gun

8.0 Acknowledgements

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